SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Sudharsanan Narashiman	**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I Sudharsanan Narashiman/192524061 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

SMART PET CARE AND VETERINARY MANAGEMENT PLATFORM

CO1 Assessment Tool 2 – Agile Sprint Planning & User Story Workshop

Student Name: Sudharsanan Narashiman
Course: Software Engineering | Course Code: CSA10 / CSA1005

1. Project Vision and Stakeholder Identification

1.1 Project Vision

The vision of the Smart Pet Care and Veterinary Management Platform is to provide a single, reliable and easy-to-use web application that helps pet owners organize their pets' profiles, vaccination histories, medical records, appointments, medication schedules and grooming activities. The platform will also support authorized veterinary professionals in reviewing and updating relevant records. By bringing these activities together, the system aims to reduce missed preventive-care tasks, improve communication between pet owners and clinics, and make pet-care information easier to access.

1.2 Objectives

- Create and maintain accurate pet profiles, including name, species, breed, age, sex and owner details.
- Store vaccination histories and upcoming vaccination due dates.
- Schedule, view and manage veterinary appointments.
- Maintain medical records and consultation notes entered by authorized veterinary professionals.
- Generate reminders for appointments, vaccinations, medication and routine preventive care.
- Track medication name, dosage instructions, frequency, start date and end date.
- Plan and track grooming and other routine-care activities.
- Provide a clear communication channel between pet owners and veterinary clinics.
- Generate useful pet-care summaries and reports.

1.3 Key Stakeholders and Their Responsibilities

Stakeholder	Role / Interest	Main Responsibilities
Pet owner	Primary user of the platform	Register pets, view records, request appointments, manage reminders and review care reports.
Veterinarian	Clinical user	Review pet history, record consultations, update vaccination and medical records, and advise owners.
Veterinary clinic receptionist / staff	Appointment support	View appointment requests, confirm or reschedule visits, and maintain clinic scheduling information.
System administrator	Platform operation	Manage user access, roles, basic configuration, backups and system availability.
Development team	Build and maintain the product	Design, implement, test, deploy and improve the web platform.
Pet / animal	Beneficiary of the service	Receives timely preventive care, treatment and grooming; represented through its records.

1.4 Project Scope

The initial release will focus on pet profiles, vaccination records, appointments, basic medical records, medication schedules, care reminders and grooming schedules. Role-based access will distinguish pet owners from veterinary staff. Online payments, video consultations, automatic diagnosis and integrations with external clinic systems are outside the first sprint and may be considered in later releases.

2. Epics and User Stories

An epic is a broad area of functionality that can be divided into smaller user stories. Each story below follows the required Agile format: "As a [user], I want [feature], so that [benefit]." The stories are written from the perspective of users who will benefit from the feature.

Epic ID / Epic	Story ID	User Story	Priority
E1 – Pet Profile Management	US-01	As a pet owner, I want to create a pet profile so that the pet's basic information is stored in one place.	Must
E1 – Pet Profile Management	US-02	As a pet owner, I want to edit a pet's profile so that its details remain accurate when they change.	Should
E2 – Vaccination Management	US-03	As a pet owner, I want to record vaccinations and due dates so that I can track preventive care.	Must
E3 – Appointment Management	US-04	As a pet owner, I want to request a veterinary appointment so that I can arrange a clinic visit.	Must
E3 – Appointment Management	US-05	As clinic staff, I want to confirm or reschedule appointment requests so that clinic visits can be organized.	Should
E4 – Medical Records	US-06	As a veterinarian, I want to add consultation notes to a pet's medical record so that relevant treatment history is maintained.	Must
E5 – Care Reminders	US-07	As a pet owner, I want to see upcoming care reminders so that I do not miss important preventive-care activities.	Must
E6 – Medication Tracking	US-08	As a pet owner, I want to record medication instructions and dates so that I can follow the prescribed schedule.	Should
E7 – Grooming & Routine Care	US-09	As a pet owner, I want to schedule grooming activities so that routine care is planned.	Could
E8 – Communication	US-10	As a pet owner, I want to send a message to the clinic so that I can clarify care or appointment details.	Could
E9 – Reports	US-11	As a pet owner, I want a summary of my pet's care history so that I can review records when needed.	Could
E10 – Access Control	US-12	As an administrator, I want to assign user roles so that only authorized users can access or update protected records.	Must

3. Acceptance Criteria

Acceptance criteria define observable conditions that must be satisfied before a user story is accepted. The following Given–When–Then criteria are measurable and can be tested.

US-01 – Create pet profile

- Given a signed-in pet owner, when they enter all required pet details and save, then the system creates the profile and displays it in their pet list.
- Given a required field is missing, when the owner tries to save, then the system identifies the missing field and does not create an incomplete profile.
- Given the profile is saved successfully, then the profile has a unique record ID and is associated with the signed-in owner.

US-03 – Record vaccination

- Given an authorized owner or veterinary user opens a pet profile, when they enter the vaccine name, date administered and next due date where applicable, then the vaccination record is saved.
- Given a due date is entered earlier than the administration date, when the user saves, then the system displays a validation message.
- Given a vaccination due date is approaching, then the reminder list displays the vaccine and its due date.

US-04 – Request appointment

- Given an owner has selected a pet, when they choose an available date/time and submit a request, then the system stores the request with a pending status.
- Given required appointment details are missing, when the owner submits, then the request is not saved and the missing details are indicated.
- Given a request is saved, then the owner can view its date, time, reason and current status.

US-06 – Add medical record

- Given a signed-in veterinarian has permission to access the selected pet, when they enter consultation date, notes and treatment details and save, then the record is attached to that pet's history.
- Given a user does not have veterinary permission, when they attempt to add or edit a clinical record, then the system denies the change.
- Given a record is saved, then its author and creation date are recorded for accountability.

US-07 – View care reminders

- Given a pet has a future vaccination, appointment or preventive-care due date, when the owner opens the reminder page, then the item appears with its due date and status.
- Given a reminder date has passed, then the item is marked overdue or past due rather than being shown as upcoming.
- Given there are no scheduled items, then the page displays a clear message that no upcoming reminders are available.

US-08 – Track medication

- Given an owner enters the medication name, prescribed instructions and schedule dates, when the information is saved, then it appears in the pet's medication list.
- Given the end date is before the start date, when the user saves, then the system rejects the invalid date range.
- Given a medication schedule is active, then its next scheduled care item is visible to the owner.

US-12 – Role-based access

- Given a pet owner is signed in, when they access the platform, then they can manage their own pet profiles but cannot edit veterinarian-only clinical notes.
- Given a veterinarian is signed in and authorized, when they open a permitted pet record, then they can add relevant clinical information.
- Given an unauthenticated user attempts to access a protected page, then the system requires sign-in before displaying protected information.

4. Product Backlog Prioritization

The product backlog is ordered by user value, risk reduction and dependencies. Must-have items are needed for a useful and safe first release. Pet profiles should be implemented before other features because vaccinations, appointments, medication and medical records must be linked to a specific pet. Role-based access is also a foundational requirement because veterinary information should only be updated by authorized users.

Rank	Story	Priority	Reason / Dependency
1	US-12 – Role-based access	Must	Protects private records and defines what each user can do.

2	US-01 – Create pet profile	Must	Core data record required by most other features.
3	US-03 – Record vaccination	Must	Directly supports preventive care and due-date tracking; depends on pet profiles.
4	US-04 – Request appointment	Must	Core clinic workflow; depends on a pet profile.
5	US-06 – Add medical record	Must	Maintains clinical history; requires role checks and a pet profile.
6	US-07 – View care reminders	Must	Combines due dates from vaccination and appointment information.
7	US-02 – Edit pet profile	Should	Improves accuracy after initial profile creation.
8	US-05 – Confirm/reschedule appointment	Should	Improves clinic workflow after appointment requests work.
9	US-08 – Track medication	Should	Adds medication scheduling after the basic pet record exists.
10	US-09 – Grooming schedule	Could	Useful routine-care feature but not required for the core first release.
11	US-10 – Clinic messaging	Could	Improves communication; can follow core records and appointments.
12	US-11 – Care reports	Could	Useful summary feature after enough records are available.

Prioritization method: MoSCoW (Must, Should, Could). This is a proposed backlog for the assessment and can be refined with feedback from pet owners and veterinary staff.

5. Sprint Backlog and Task Breakdown

5.1 Sprint Planning Assumptions

Sprint duration: 2 weeks. Team assumption: one beginner-level full-stack developer or a small student team, with a reviewer acting as the product owner. The sprint focuses on a small but demonstrable foundation: sign-in/role checks, pet profile creation, vaccination entry and a basic list of care reminders. Actual velocity should be recalculated after the team's first sprint.

5.2 Selected Sprint 1 Stories

Story	Sprint selection	Reason
US-12 – Role-based access	Selected	Access control is needed before private pet records are exposed.
US-01 – Create pet profile	Selected	Provides the central record used by later features.
US-03 – Record vaccination	Selected	Demonstrates preventive-care record keeping.
US-07 – View care reminders (basic)	Selected in limited form	Shows upcoming vaccination due dates from saved data.
US-04 – Appointment request	Next sprint	Kept for the next sprint if the initial team capacity is limited.

5.3 Actionable Development Tasks

Task ID	Development task	Expected result
T1	Confirm required fields and screen sketches for login, pet profile, vaccination and reminders.	Agreed mini-requirements and simple UI design.
T2	Create database tables for users, pets and vaccinations with primary and foreign keys.	Pet and vaccination data can be stored consistently.

T3	Implement sign-in and role checks for owner, veterinarian and administrator roles.	Protected pages require authentication and permissions.
T4	Build the create-pet form with required-field validation.	Owner can add a pet profile and see it in the pet list.
T5	Build vaccination entry and history display linked to a pet.	Vaccination details can be added and viewed.
T6	Build a reminder view that lists upcoming vaccination due dates.	Owner can see upcoming due dates and overdue items.
T7	Test valid/invalid forms, user permissions, record links and date handling.	Defects are identified and corrected.
T8	Prepare sample data, demonstration steps and a short sprint review.	Working demo and review notes are ready.

5.4 Definition of Done

- The feature matches its acceptance criteria.
- Required fields and invalid dates are validated.
- Protected actions check the signed-in user's role and ownership/permission.
- The feature has been tested using normal and error scenarios.
- No known critical defect remains in the sprint scope.
- The code and setup notes are saved, and the feature can be demonstrated.

6. Story Point Estimation and Justification

Story points estimate relative effort, complexity and uncertainty rather than exact hours. The Fibonacci sequence (1, 2, 3, 5, 8, 13) is used. The estimates below are proposed planning estimates; a real team would agree them through Planning Poker.

Story	Points	Justification
US-12 – Role-based access	5	Requires authentication checks, role rules and negative permission tests.
US-01 – Create pet profile	3	A basic form, validation, database save and list view are moderately straightforward.
US-03 – Record vaccination	5	Includes linked records, date validation, history display and due-date handling.
US-07 – Basic care reminders	3	Requires selecting future/overdue due dates and displaying them clearly.
Total Sprint 1	16	A tentative target for a small team; reduce scope if the team's actual capacity is lower.

Estimation scale: 1 = very small and well understood; 2 = small; 3 = moderate; 5 = complex or involving several components; 8 = large with notable uncertainty; 13 = too large and should normally be split. The total of 16 points is not a guarantee of completion. It is a forecast that should be checked against team availability and previous velocity.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

By the end of Sprint 1, a signed-in pet owner should be able to create and view a pet profile, record a vaccination, and see upcoming vaccination reminders, while role checks prevent unauthorized users from changing protected records.

7.2 Expected Deliverables

- A basic web application with sign-in and role-based access checks.
- A pet profile form and a list of saved pet profiles.
- A vaccination entry form and vaccination history linked to each pet.

- A reminder page showing upcoming and overdue vaccination due dates.
- Database structure for users, pets and vaccination records.
- Test results for successful entry, missing fields, invalid dates and unauthorized access.
- A short setup guide, sample data and a working demonstration.

7.3 Sprint Review and Demonstration Plan

1. Sign in as a pet owner using a demonstration account.
2. Create a sample pet profile and show that it appears in the pet list.
3. Open the pet profile and add a vaccination record with a due date.
4. Open the reminder page and show the upcoming vaccination.
5. Try to perform a protected action with a user role that lacks permission and show that access is denied.
6. Review the acceptance criteria with the product owner and record any feedback as new backlog items.

8. Risks, Assumptions and Future Improvements

Risk / Assumption	Response
Incorrect or incomplete pet information	Validate required fields and allow authorized corrections.
Unauthorized access to medical information	Use authentication, role-based authorization and ownership checks; test denied access.
Reminders may not be delivered automatically in the first version	Start with an in-app reminder list; add email/SMS/push delivery in a later sprint.
Team capacity is uncertain	Use the sprint review to compare planned and completed points and adjust future commitments.
Clinical information needs professional oversight	Only authorized veterinary professionals should create or update veterinary consultation records.

Future improvements may include appointment confirmation and rescheduling, medication reminders, grooming calendars, secure clinic messaging, downloadable care reports, email or mobile notifications, and integration with veterinary clinic systems. Any future feature should be prioritized with stakeholder feedback.

9. Conclusion

The proposed Agile plan converts the Smart Pet Care and Veterinary Management Platform problem statement into a clear product vision, stakeholder list, epics, user stories, testable acceptance criteria and a prioritized backlog. Sprint 1 concentrates on the foundation needed for later features: role-based access, pet profiles, vaccination records and basic reminders. Using a two-week sprint, task breakdown, Fibonacci story points and a defined sprint goal gives the team a practical way to build, test and demonstrate progress. The backlog can then be refined in later sprints to add appointments, medical records, medication tracking, grooming, communication and reports.